Lecture 3: Linear dimensionality reduction, continued

TTIC 31220: Unsupervised Learning and Data Analysis

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TTI-Chicago

March 29, 2026

Announcements

- Tutorial 2 this Wednesday: Kernels
- Homework 1 released, due April 9 5:59pm
- (Tentative) timeline of assignment deadlines and quizzes is on syllabus page
- Small update to readings for PCA

An intro to Homework 1...

Content

- PCA, CCA, probabilistic PCA (some material to be covered this week)
- Thinking about hyperparameters

Other notes

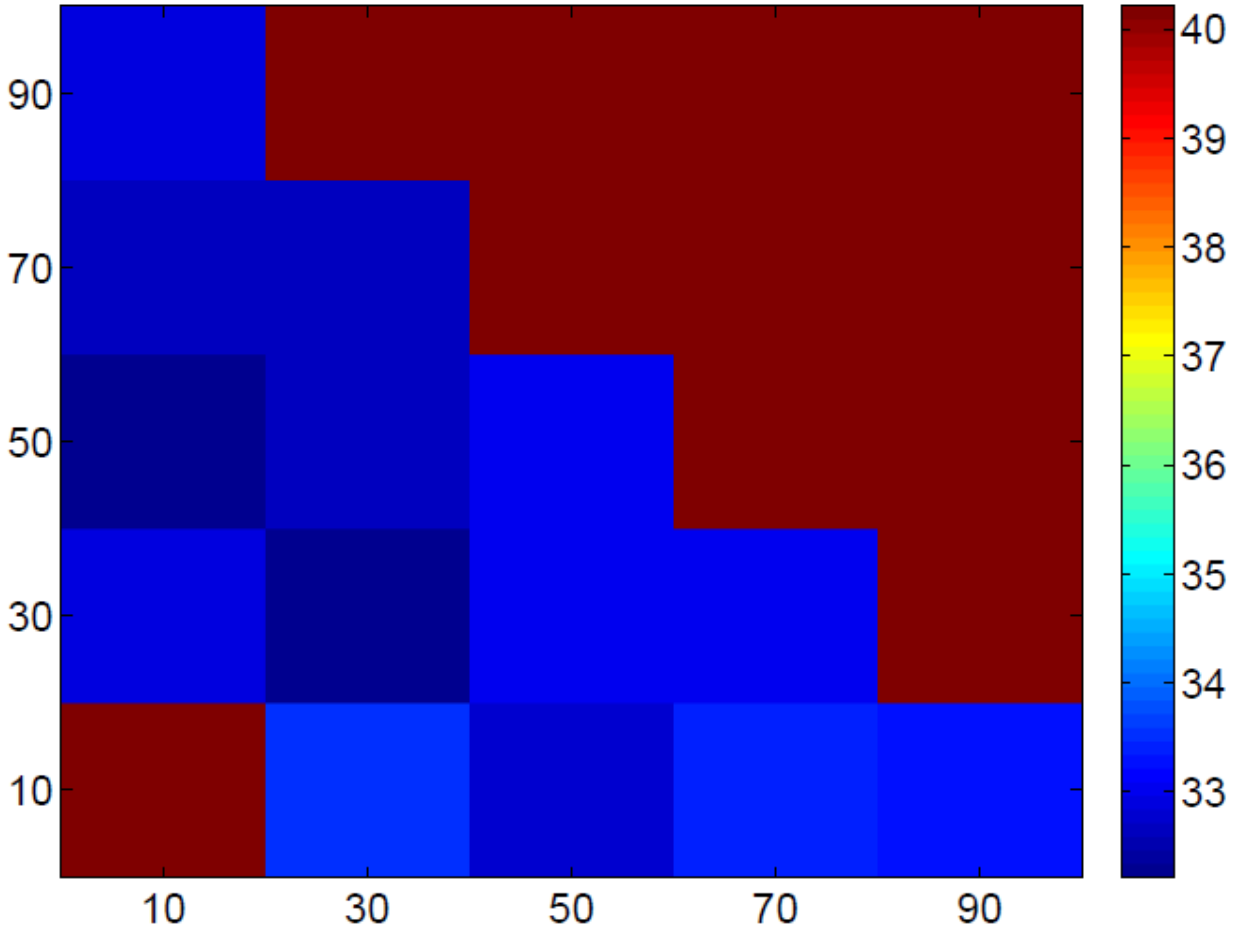
- Please review the lateness policy and the rest of the long preamble (it will be the same for future homeworks!)
- Try things out sooner rather than later
- Don't forget to submit answers to question 1 (feedback)!

Practical ML: Hyperparameter tuning

Tuning is the process of finding the sweet spot in the "heat map" of performance vs. hyperparameter values:

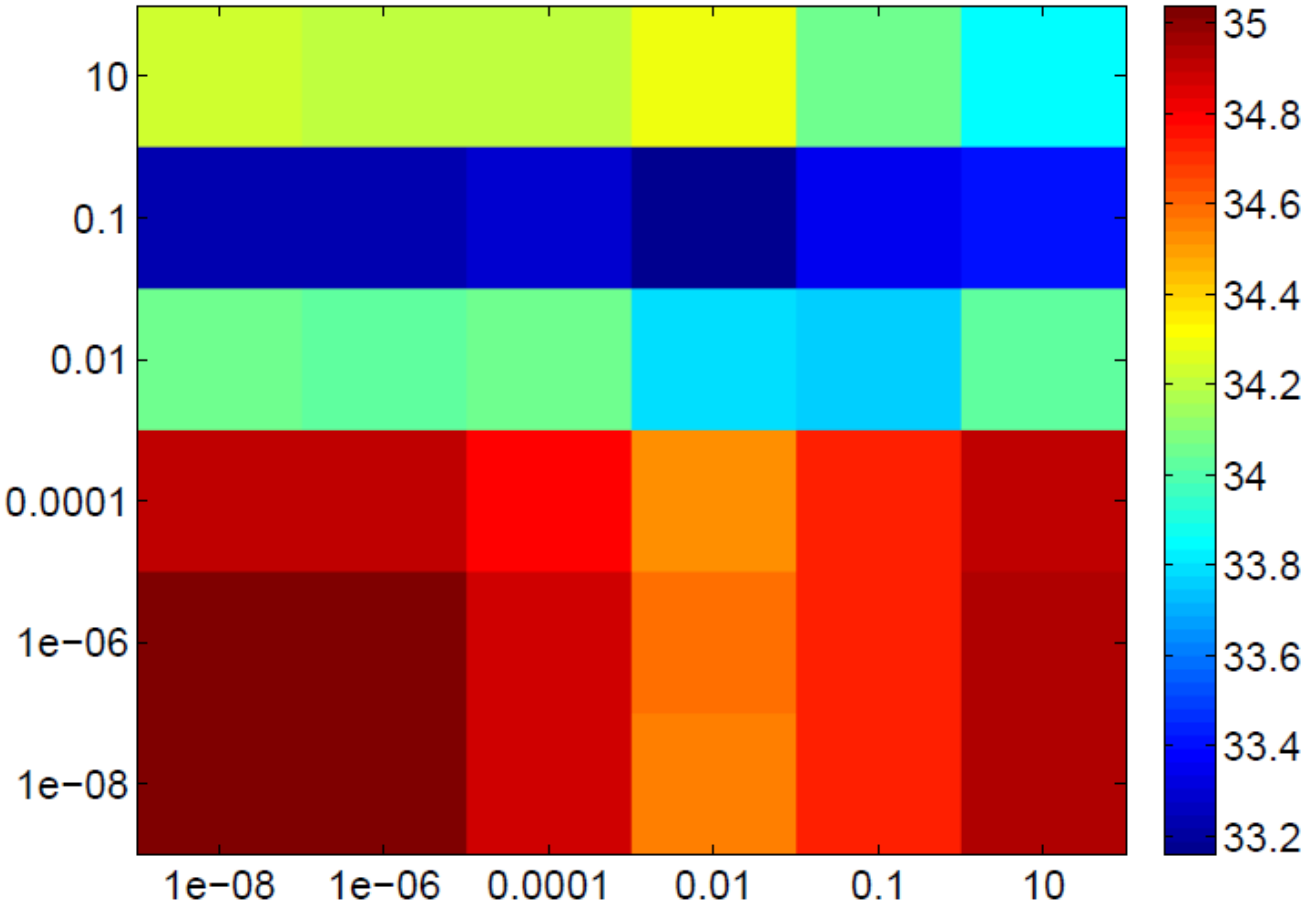
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Practical ML: Hyperparameter tuning

Tuning is the process of finding the sweet spot in the "heat map" of performance vs. hyperparameter values:



The ML practitioner's plain vanilla experimental loop

- **Think** of good initial values for hyperparameters
- **Train and test** with a coarse "grid" of hyperparameters including the ones you think are likely to work well
- **Examine results** (via a heat map or otherwise) and, possibly, come up with a new set of hyperparameters on a finer grid
  - Previous best results on the edge of the grid? Then **expand** in that direction
  - Previous best results in a small part of the coarse grid? Then use a finer grid to **zoom in** within that region

The ML practitioner's plain vanilla experimental loop

Other notes:

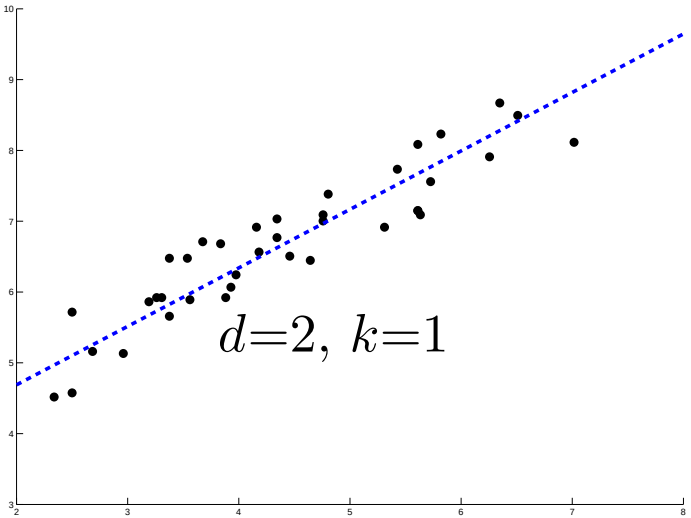
- Alternative techniques include gradient descent-like hyperparameter optimization
- Think ahead of time how long a set of tuning experiments will take you. If you have a grid of $10 \times 10 \times 10$ hyperparameter values that you want to try out, and each setting will take you 1 minute to get a result...
- Don't try *too hard* on the tuning set. If the sweet spot of performance is in a tiny part of the heat map, then it may not generalize to the test set.
- How hard we need to work at tuning is one measure of the usefulness of a technique!

Representation learning: Linear dimensionality reduction

- Principal component analysis (PCA) (Last time)
- Probabilistic principal component analysis (PPCA)
- Canonical correlation analysis (CCA, started last time)
- Multidimensional scaling (MDS)

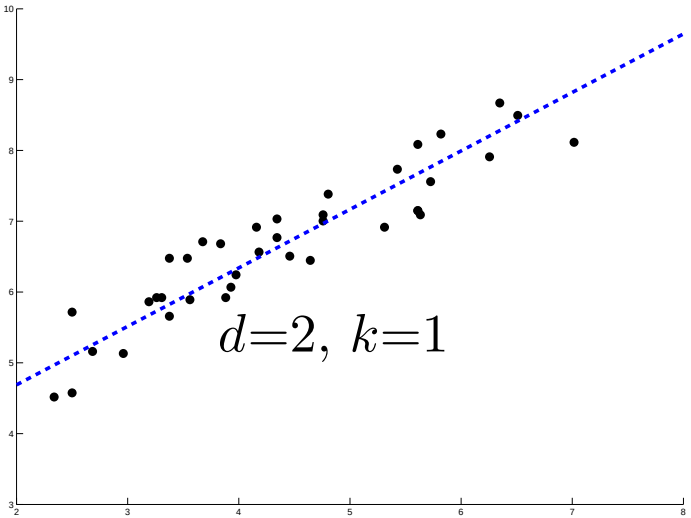
Dimensionality reduction

- d: the "nominal" or "ambient" dimensionality
- There is often a lower *intrinsic* dimensionality: a property of the generating process $\Rightarrow$ the data lie on (or near) a subspace of dimensionality $k < d$.



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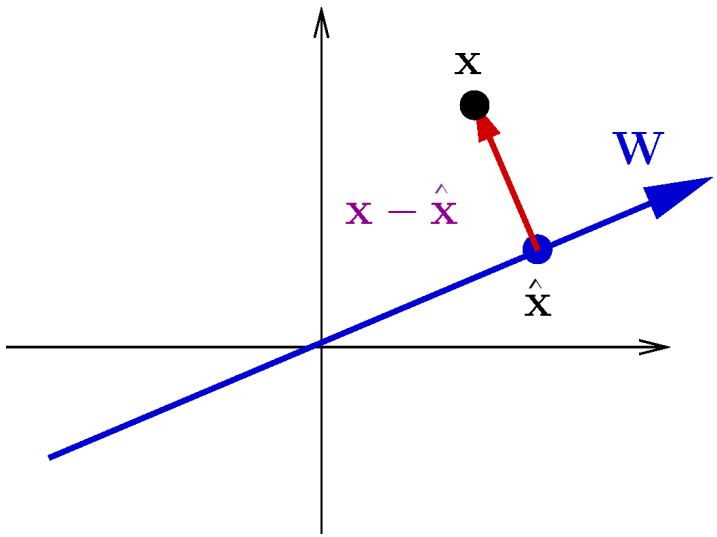
Dimensionality reduction seeks to recover simple structure that "explains" the data

- by finding an underlying subspace
- equivalently, by recovering hidden variables (directions of variation)

Linear dimensionality reduction

- Assume for now that the data points have zero mean
- Linear subspace is defined by a $d \times k$ matrix $\mathbf{W}$ containing k direction vectors
- Approximation (reconstruction) of $\mathbf{x}$ as a weighted combination of the vectors in $\mathbf{W}$:

$$\hat{\mathbf{x}} = \mathbf{W}\mathbf{z}$$



- $\mathbf{z}$ is a data point-specific vector of k weights
- $\mathbf{z}$ can be seen as a low-dimensional representation of $\mathbf{x}$

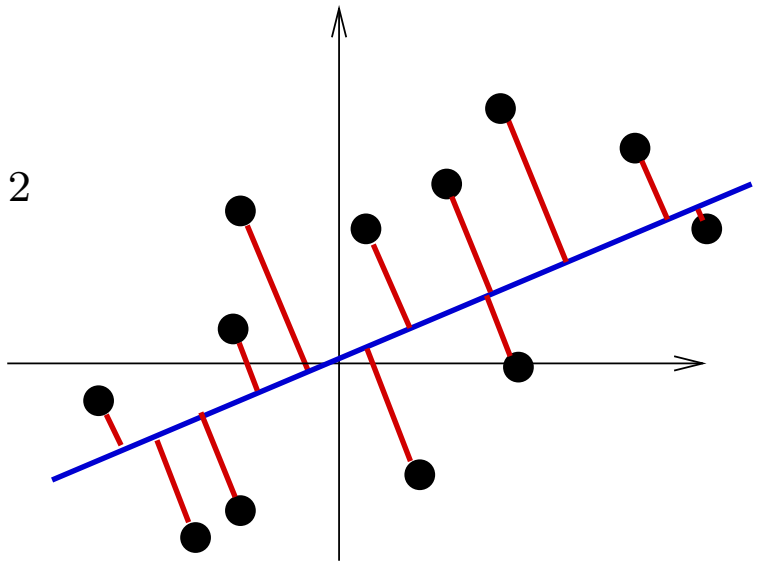
Principal component(s) analysis (PCA)

Proposed by Pearson (1901) and further (independently?) developed by Hotelling in 1933

- A linear dimensionality reduction technique
- **Objective**: minimize the sum of *residuals* subject to orthogonality of projection directions (= "principal directions")

$$\mathbf{W}^* = \underset{\mathbf{W}}{\operatorname{argmin}} \sum_{i=1}^{N} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$$

$$\text{s.t. } \mathbf{W}^T\mathbf{W} = I$$



Principal component(s) analysis (PCA)

PCA is the solution to the minimum-residual projection problem, which is as follows:

- Let the columns of $\mathbf{X}$ be the data points ($d \times N$ matrix)
- Construct the $d \times d$ *data covariance matrix* $\mathbf{S} = \frac{1}{N}\mathbf{X}\mathbf{X}^T$;
- Let $\phi_1, \ldots, \phi_d$ be the orthonormal eigenvectors of $\mathbf{S}$ corresponding to the eigenvalues $\lambda_1 \geq \ldots \geq \lambda_d$.

$$\phi_l^T \phi_j = \begin{cases} 0 & \text{if } l \neq j, \\ 1 & \text{for } l = j. \end{cases}$$

- Then the optimal k-dim linear subspace turns out to be given by

$$\mathbf{W} = \mathbf{\Phi} = [\phi_1, \ldots, \phi_k].$$

PCA: maximum variance subspace formulation

- An alternative criterion: maximize the *variance* of the projection

$$\begin{aligned} \mathbf{W}^* &= \underset{\mathbf{W}}{\operatorname{argmax}} \operatorname{tr}\left(\mathbf{W}^T \mathbf{X}\mathbf{X}^T \mathbf{W}\right) \\ \text{s.t. } \mathbf{W}^T\mathbf{W} &= I \end{aligned}$$

- Leads to the same solution as the minimum-residual formulation
- We (partially) proved this claim, and derived the $k = 1$ solution

Probabilistic PCA (PPCA)

Both a generalization of PCA and a probabilistic interpretation of it

- Independently proposed by [Tipping & Bishop 1997, 1999] and [Roweis 1998]

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Model: $\mathbf{z} \in \mathbb{R}^k, \mathbf{x} \in \mathbb{R}^d, k \leq d$

$$\begin{aligned} p(\mathbf{z}) &= \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I}) \\ p(\mathbf{x}|\mathbf{z}) &= \mathcal{N}(\mathbf{x}; \mathbf{W}\mathbf{z} + \mu, \sigma^2\mathbf{I}) \end{aligned}$$

In other words:

$$\mathbf{x} = \mathbf{W}\mathbf{z} + \mu + \epsilon, \text{ where } \epsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I}) \text{ and } \epsilon, \mathbf{z} \text{ are independent}$$

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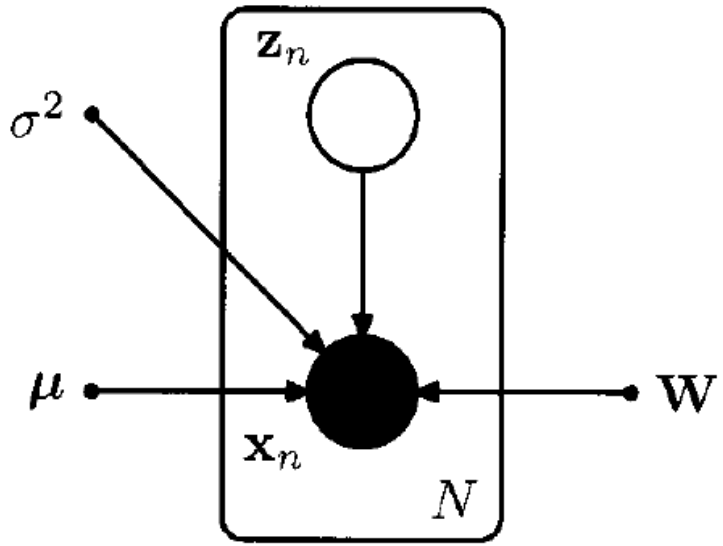
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Goal: Given N examples of $\mathbf{x}$, determine the maximum likelihood (ML) solution for $\mathbf{W}, \mu, \sigma$, i.e. that maximize the (log) likelihood $\sum_{i=1}^{N} \log p(\mathbf{x}_i)$

Probabilistic PCA (PPCA)

Graphical model representation of the PPCA model:



[Bishop Fig. 12.13]

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Marginal distribution over $\mathbf{x}$ (marginal of a Gaussian is also Gaussian):

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu, \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}) \text{ since } \mathbf{z}, \epsilon \text{ are independent}$$

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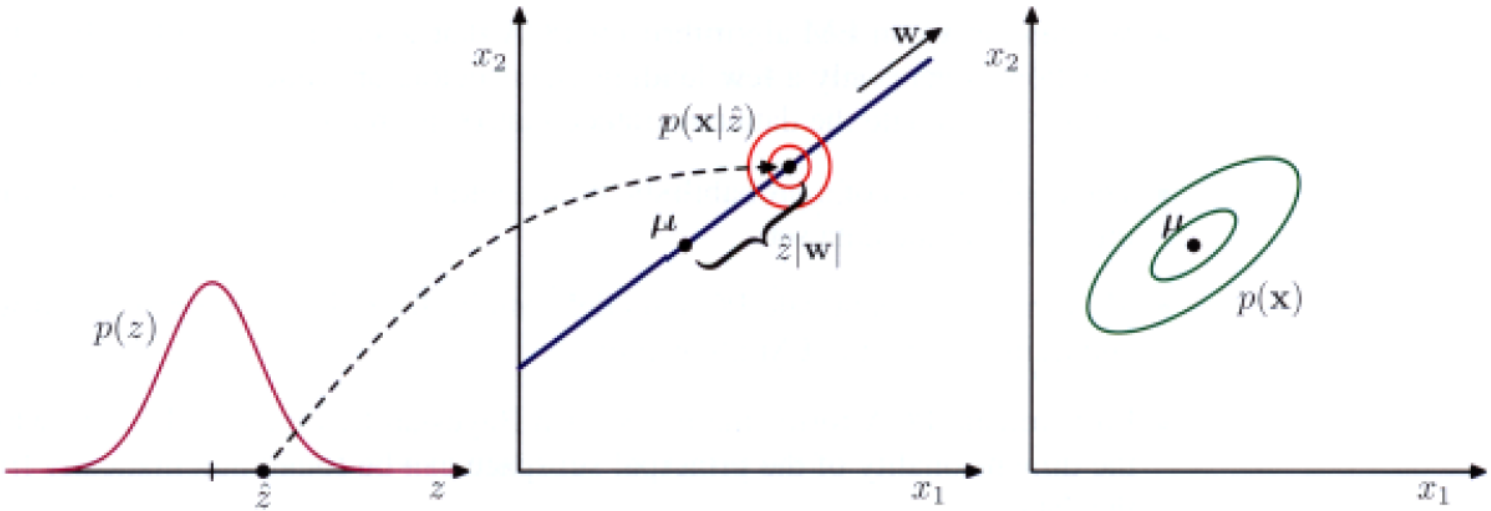
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[Bishop Fig. 12.9]

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\implies p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}; \mu, \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I})
\end{aligned}
$$

Before seeking the ML solution, notice that the solution is not unique: if $\mathbf{R}$ is an orthogonal matrix (a rotation matrix), then $\tilde{\mathbf{W}} = \mathbf{W}\mathbf{R}$ is indistinguishable from $\mathbf{W}$:

$$\tilde{\mathbf{W}}\tilde{\mathbf{W}}^T = \mathbf{W}\mathbf{R}\mathbf{R}^T\mathbf{W}^T = \mathbf{W}\mathbf{W}^T$$

So we will find a solution up to a rotation $\mathbf{R}$.

PPCA solution

$$
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\implies p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}; \mu, \mathbf{C}),\ \text{where } \mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I} \\
\implies p(\mathbf{z}|\mathbf{x}) &= \mathcal{N}(\mathbf{z}; \mathbf{M}^{-1}\mathbf{W}^T(\mathbf{x} - \mu), \sigma^2\mathbf{M}^{-1}),\ \text{where } \mathbf{M} = \mathbf{W}^T\mathbf{W} + \sigma^2\mathbf{I}
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$$

(For the last equality, see Bishop eq. 2.116)

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$$

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To solve, given a data set $\mathbf{X} = \{\mathbf{x}_i\}$, we write down the log likelihood function $\sum_{i=1}^{N} \log p(\mathbf{x}_i; \mathbf{W}, \mu, \sigma^2)$:

$$
-\frac{Nd}{2}\log(2\pi) - \frac{N}{2}\log|\mathbf{C}| - \frac{1}{2}\sum_{i=1}^{N}(\mathbf{x}_i - \mu)^T\mathbf{C}^{-1}(\mathbf{x}_i - \mu)
$$

Setting the derivative w.r.t. μ to 0, we get $\mu = \bar{\mathbf{x}}$, the data mean (see HW1).

PPCA solution

Solution w.r.t. $\mathbf{W}, \sigma^2$ is more complicated, but closed-form:

$$\begin{aligned} \mathbf{W}_{ML} &= \mathbf{U}_k(\Lambda_k - \sigma^2_{ML}\mathbf{I})^{1/2}\mathbf{R} \\ \sigma^2_{ML} &= \frac{1}{d-k}\sum_{j=k+1}^{d}\lambda_j \end{aligned}$$

where:

- $\mathbf{U}_k$: $d \times k$, columns = first k eigenvectors of the data covariance $\mathbf{S}$
- Λ_k : $k \times k$ diagonal matrix with corresponding e-values λ_j on diagonal
- $\mathbf{R}$ is an arbitrary orthogonal matrix

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Things to notice:

- For $\mathbf{R} = \mathbf{I}$, the solution for $\mathbf{W}$ is just a scaled version of that of standard PCA!
- σ^2_{ML} is the average variance of the discarded dimensions
- If $k = d$ (no dim reduction): the covariance $\mathbf{C}$ of $\mathbf{x}$ is equal to $\mathbf{S}$, the sample covariance. This is just the standard ML solution for a Gaussian distribution!

PPCA for dimensionality reduction

A reduced-dimensionality representation of $\mathbf{X}$ is given by the posterior mean of $\mathbf{z}$:

$$\mathbb{E}[\mathbf{z}|\mathbf{x}] = \mathbf{M}^{-1}\mathbf{W}_{ML}^{T}(\mathbf{x} - \bar{\mathbf{x}})$$

where $\mathbf{M} = \mathbf{W}_{ML}^{T}\mathbf{W}_{ML} + \sigma_{ML}^{2}\mathbf{I}$

- As $\sigma^2 \to 0$, the posterior mean approaches the standard PCA projection
- For $\sigma^2 > 0$, the posterior mean "shrinks" the solution. Is this a good thing?

More notes about PPCA

- It is also possible to find the PPCA solution iteratively, via the *expectation-maximization (EM)* algorithm (more on EM later in the course)
- This is useful if doing the eigendecomposition is too computationally demanding
- PPCA provides one way of approximating a Gaussian using fewer parameters
- As a probabilistic model, PPCA also provides a way of sampling from the data distribution
- PPCA was also named *sensible PCA* by Roweis

Multi-view linear dimensionality reduction

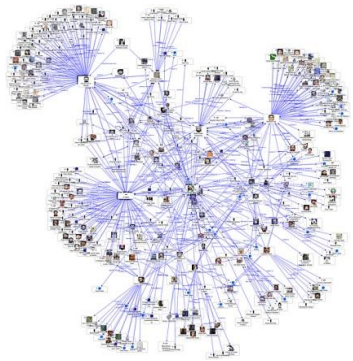
Multi-view linear dimensionality reduction

- Unsupervised dimensionality reduction is very challenging
- If we happen to have multiple data "views", it can be easier

Multi-view data examples



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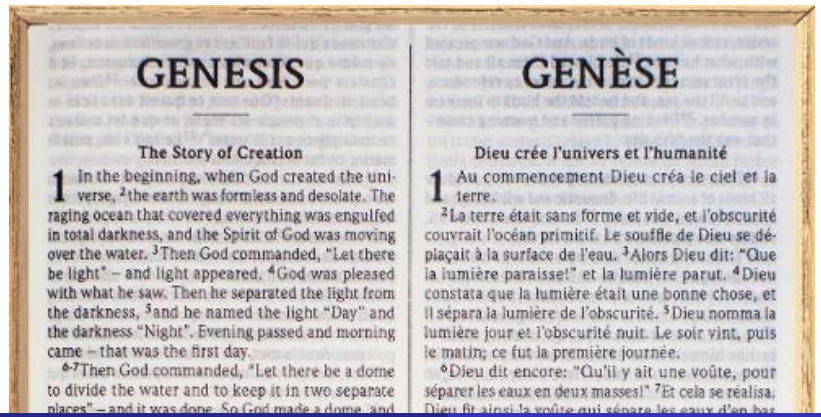


Family with 2 kids and a dog

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Several kinds of winter squash

The Earth as seen from space



GENESIS

The Story of Creation

1 In the beginning, when God created the universe, 2 the earth was formless and desolate. The raging ocean that covered everything was engulfed in total darkness, and the Spirit of God was moving over the water. 3 Then God commanded, "Let there be light" – and light appeared. 4 God was pleased with what he saw. Then he separated the light from the darkness, 5 and he named the light "Day" and the darkness "Night". Evening passed and morning came – that was the first day.

6-7 Then God commanded, "Let there be a dome to divide the water and to keep it in two separate

GENÈSE

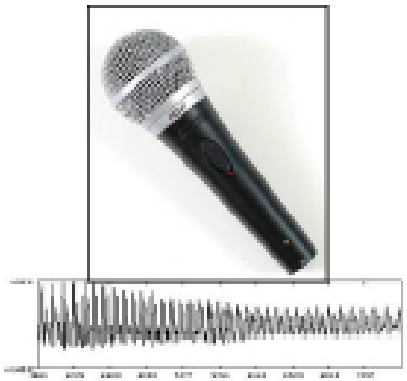
Dieu crée l'univers et l'humanité

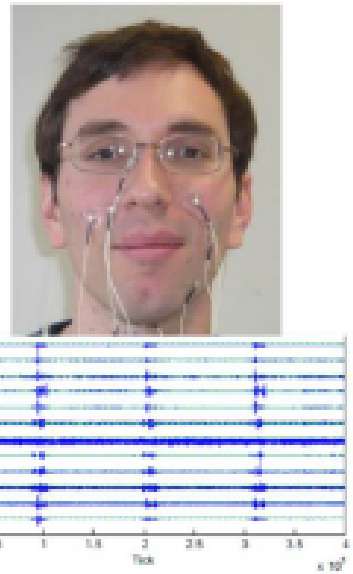
1 Au commencement Dieu créa le ciel et la terre.

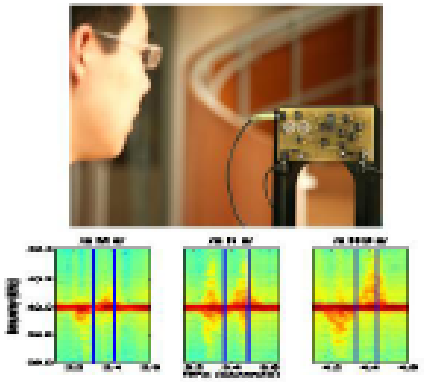
2 La terre était sans forme et vide, et l'obscurité couvrait l'océan primitif. Le souffle de Dieu se déplaçait à la surface de l'eau. 3 Alors Dieu dit: "Que la lumière paraisse!" et la lumière parut. 4 Dieu constata que la lumière était une bonne chose, et il sépara la lumière de l'obscurité. 5 Dieu nomma la lumière jour et l'obscurité nuit. Le soir vint, puis le matin; ce fut la première journée.

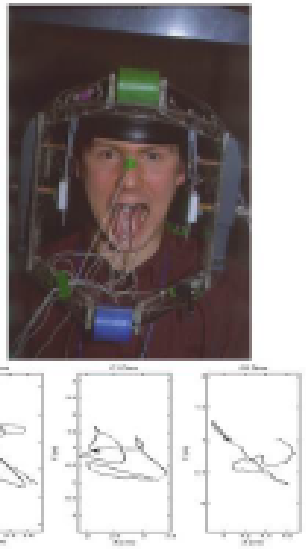
6 Dieu dit encore: "Qu'il y ait une voûte, pour séparer les eaux en deux masses!" 7 Et cela se réalisa.

Multi-view data examples: Speech

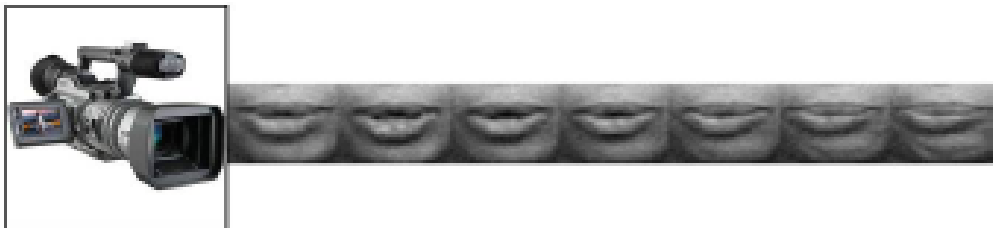








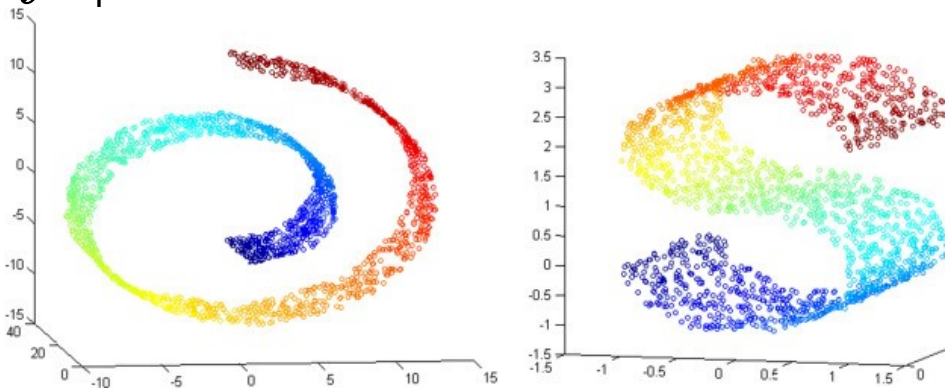




Multi-view representation learning (linear or not)

Training data consists of samples of a d-dimensional random vector that has some natural split into two sub-vectors:

$$\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}, \ \mathbf{x} \in \mathbf{R}^{d_x}, \ \mathbf{y} \in \mathbf{R}^{d_y}, \ d_x + d_y = d$$



- **Multi-view representation learning**: The problem of learning useful features/subspaces from such two-view data
- Typically involves learning representations of one view that are **predictive** of the other view, and vice versa
- Intuition: If the noise/nuisance parameters in the two views are independent, then the common information must be signal!

Canonical correlation analysis (CCA) [Hotelling 1936]

- Given: data set of n paired vectors $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, which are samples of random vectors $X \in \mathbf{R}^{d_x}, Y \in \mathbf{R}^{d_y}$
- Find: direction vectors v, w that maximize the correlation between the projections $v^T X$ and $w^T Y$

(note: vs and ws not orthonormal!)

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- Subsequent direction vectors maximize the same correlation, subject to being uncorrelated with the previous directions

(note: vs and ws not orthonormal!)

Canonical correlation analysis (CCA) [Hotelling 1936]

- Given: data set of n paired vectors $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, which are samples of random vectors $X \in \mathbf{R}^{d_x}, Y \in \mathbf{R}^{d_y}$
- Find: direction vectors v, w that maximize the correlation between the projections $v^T X$ and $w^T Y$
- First pair of directions:

$$\begin{aligned} v_1, w_1 &= \underset{v,w}{\operatorname{argmax}} \ \operatorname{corr}(v^T X, w^T Y) \\ &= \underset{v,w}{\operatorname{argmax}} \ \frac{v^T C_{xy} w}{\sqrt{(v^T C_{xx} v)(w^T C_{yy} w)}} \end{aligned}$$

- Subsequent direction vectors maximize the same correlation, subject to being uncorrelated with the previous directions
- Equivalent to maximizing the covariance, subject to $v^T C_{xx} v = 1, w^T C_{yy} w = 1$ (note: vs and ws not orthonormal!)

Canonical correlation analysis (CCA)

Solution can be found via an eigenproblem:

$$\begin{aligned} C_{xx}^{-1} C_{xy} C_{yy}^{-1} C_{yx} v &= \lambda^2 v \\ w &\propto C_{yy}^{-1} C_{yx} v \end{aligned}$$

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Can use CCA for dimensionality reduction of just one view, say View 1 (X):

- Compute first k CCA directions
- Project new data from View 1 onto those directions

Canonical correlation analysis (CCA)

Alternative formulations: Find $V \in \mathbf{R}^{d_x \times k}, W \in \mathbf{R}^{d_y \times k}$ that

$$(1) \qquad \text{minimize}: \quad E||W^T Y - V^T X||_F^2$$

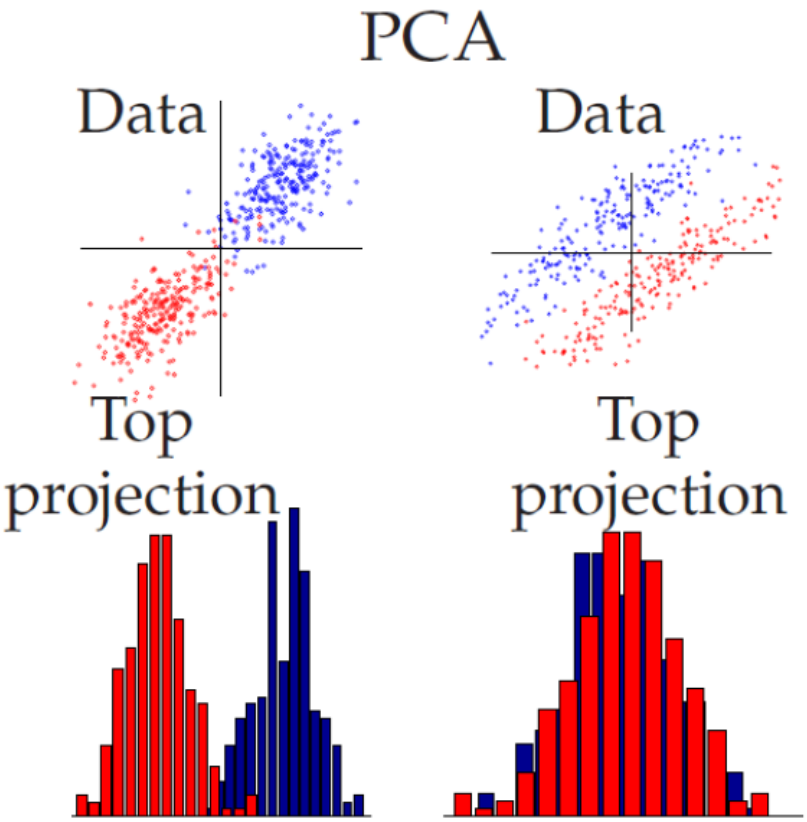
$$\text{subject to}: \quad V^T C_{xx} V = W^T C_{yy} W = I$$

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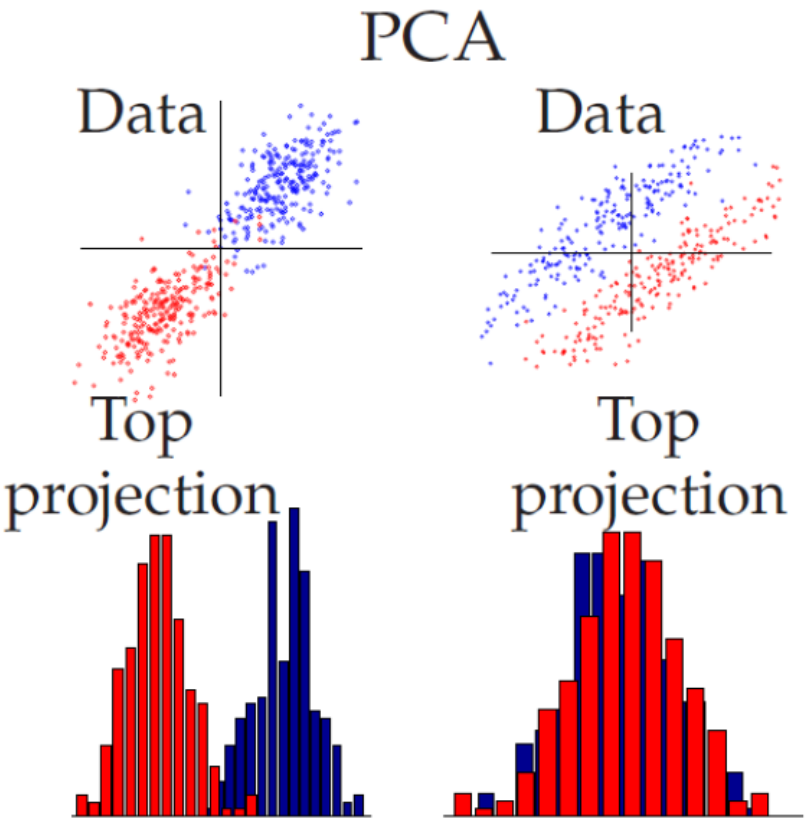
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$$\begin{aligned} & (1) && \text{minimize}: && E||W^T Y - V^T X||_F^2 \\ OR\ & (2) && \text{maximize}: && \operatorname{Tr} V^T C_{xy} W \\ & && \text{subject to}: && V^T C_{xx} V = W^T C_{yy} W = I \end{aligned}$$

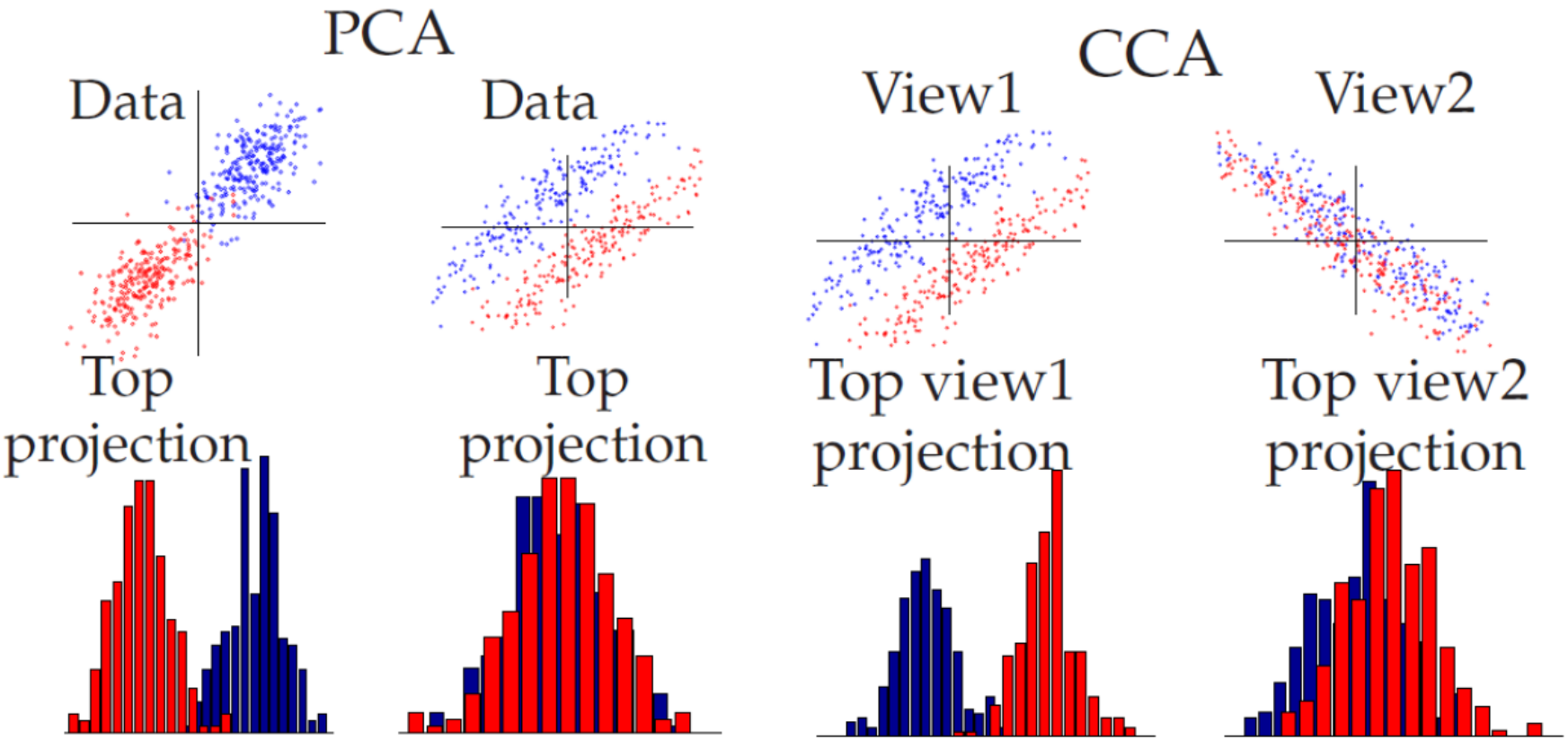
Toy examples



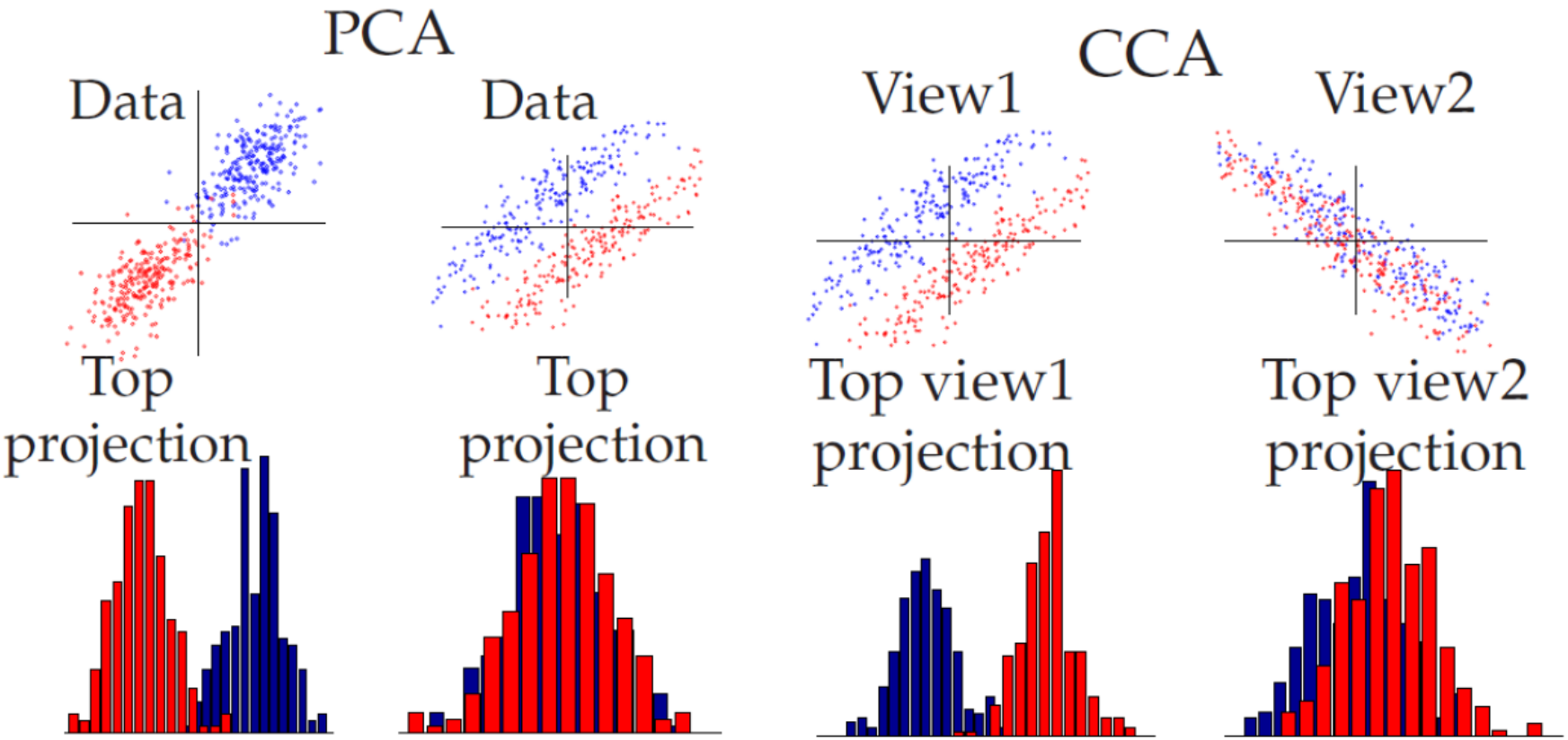
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Toy examples



- If there is a class structure in the data, CCA has discriminative properties when the two views are uncorrelated given the class label

Question from last time: CCA with > 2 views?

- No single principled extension that I know of
- Some work exists on various possible extensions, e.g.:
  Paul Horst, "Generalized canonical correlations and their applications to experimental data," *Journal of Clinical Psychology*, **17**(4), 1961.

Regularizing CCA

CCA can overfit, i.e. to find spurious correlations in the training data

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Solution becomes eigenvectors of $(C_{xx} + r_x I)^{-1} C_{xy} (C_{yy} + r_y I)^{-1} C_{yx}$

Optionally, also do an initial dimensionality reduction via PCA in each view (sometimes referred to as singular vector CCA (SVCCA))

CCA applications: Audio-visual synchronization

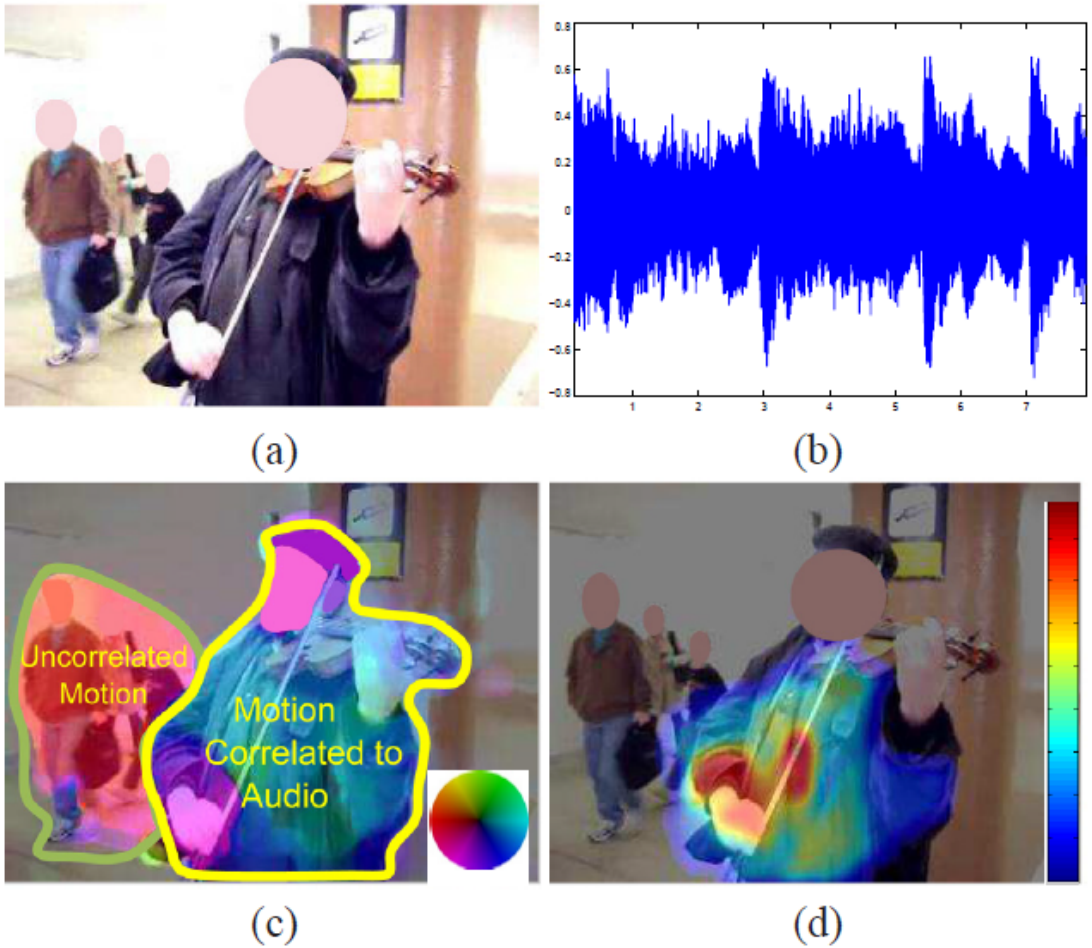


Fig. 1. (a) A sample frame of a video showing a violinist; (b) Audio signal associated with the video, shown as amplitude plot; (c) Optical flow shows the two dominant motions in the scene (flow direction shown by hue and magnitude by brightness); and (d) Localization of moving and sounding objects (the player and the violin in this case) generated by the proposed method, where the probability of correlation to the sound is shown by the colorbar.

CCA applications: Multilingual word embeddings

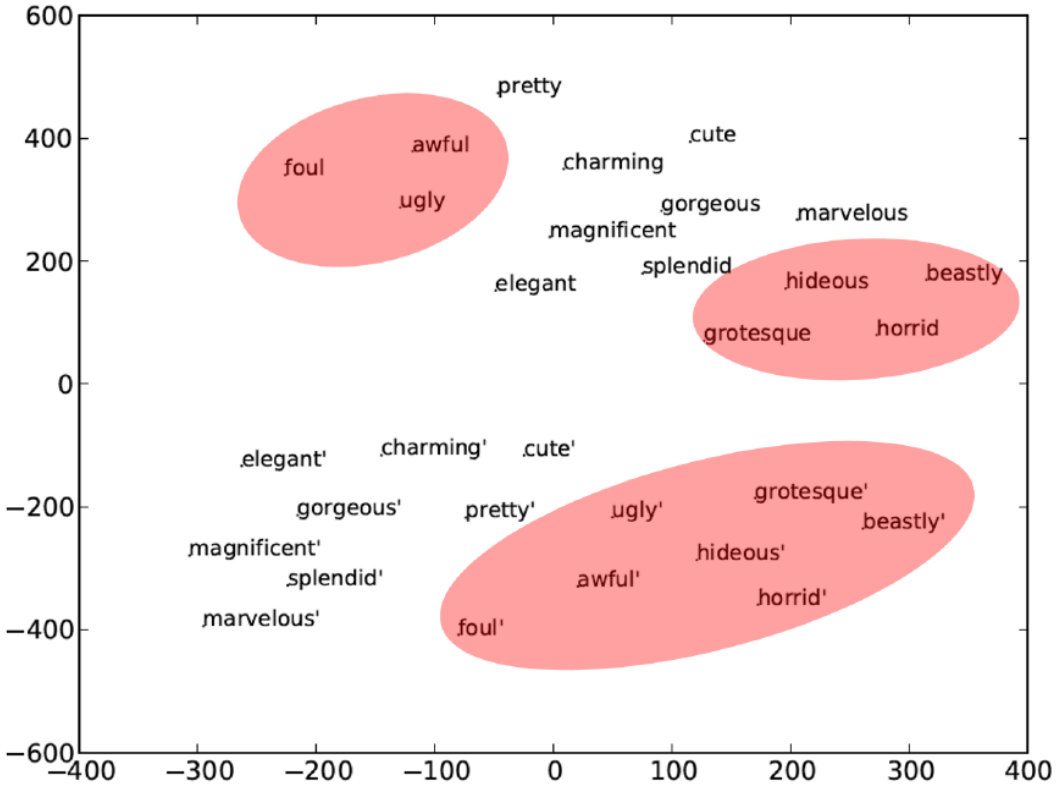
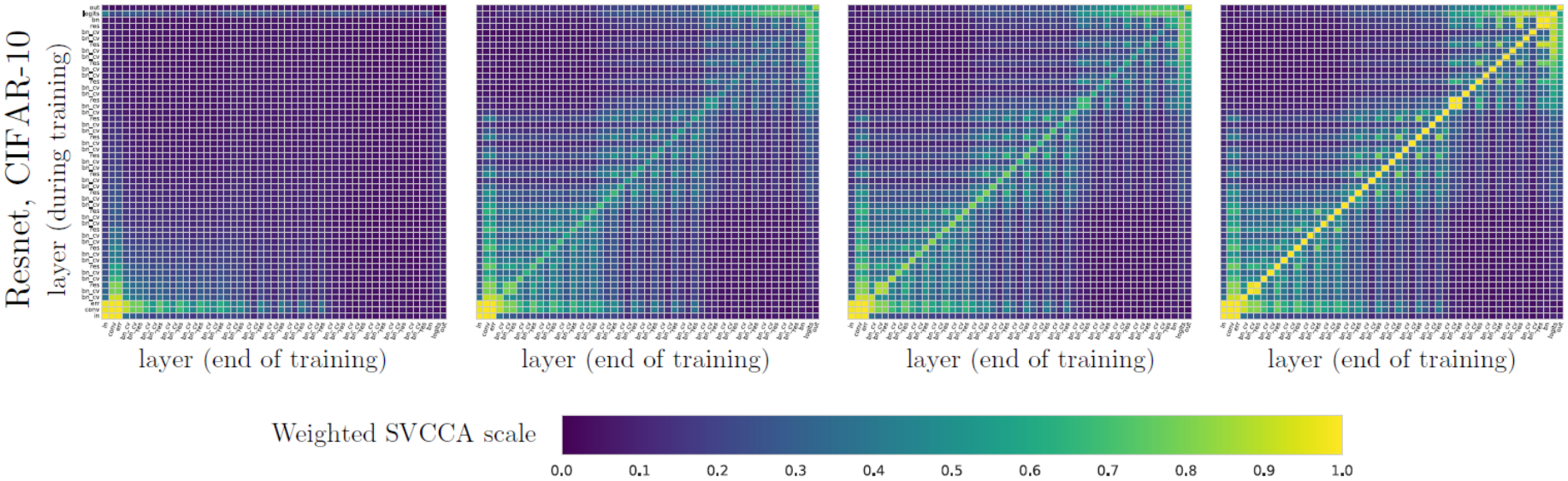


Figure 2: Monolingual (top) and multilingual (bottom; marked with apostrophe) word projections of the antonyms (shown in red) and synonyms of "beautiful".

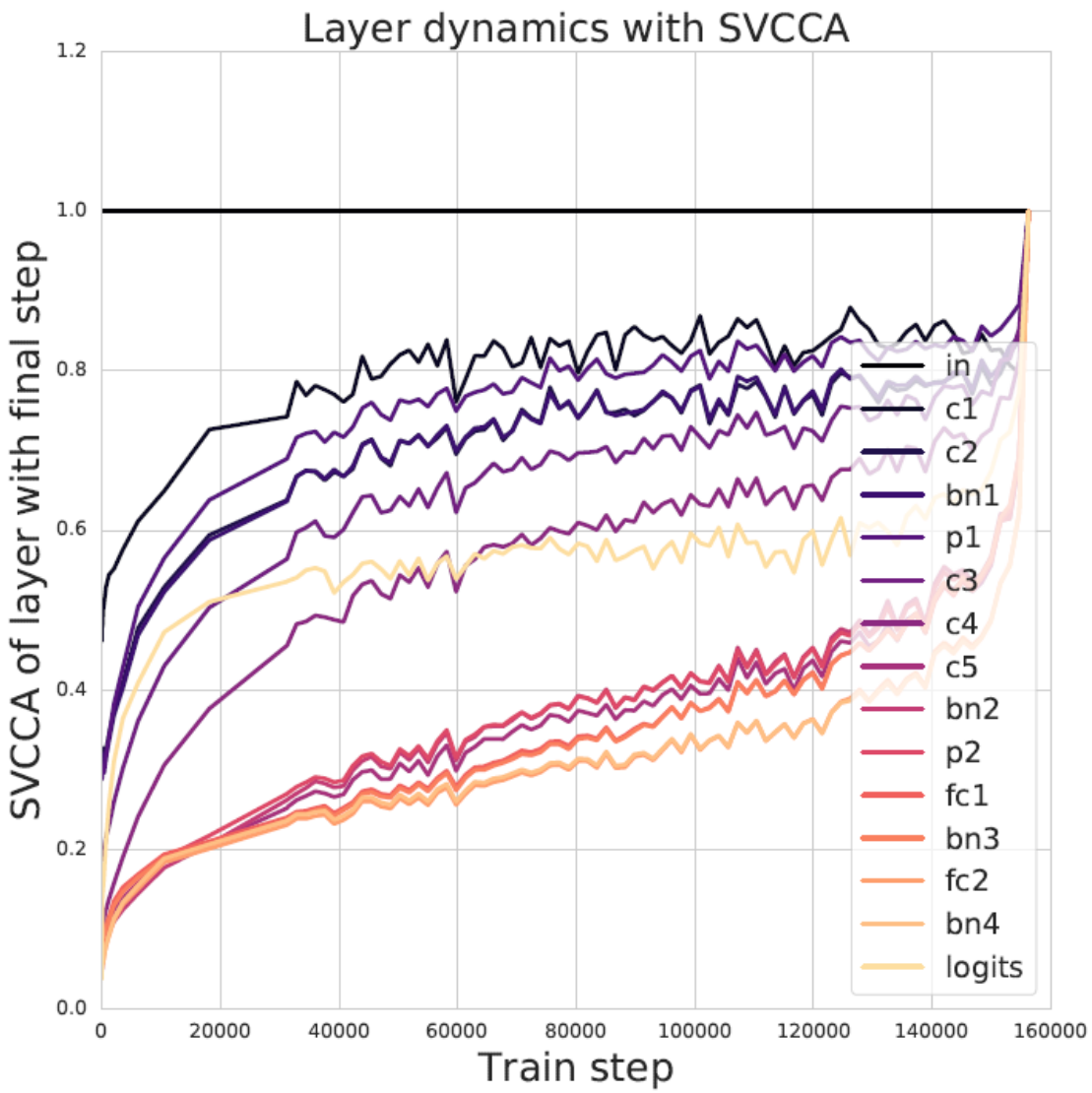
[Faruqui+ 2014]

CCA applications: Neural network analysis



[Raghu+ 2017]

CCA applications: Neural network analysis



[Raghu+ 2017]

Other multi-view dimensionality reduction applications

All the usual applications of dimensionality reduction: compression, improved classification with limited labeled data, ... but also:

- Cross-view retrieval
- To "line up" two views for synchronization/visualization/analysis

Some application domains: image-text retrieval, speech recognition, neuroscience, ...

Multidimensional scaling

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- MDS seeks a k-dimensional representation $\mathbf{z}$ of a data set that preserves inner products between pairs of data points $\mathbf{x}$:

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- The solution can be obtained from the $N \times N$ Gram matrix of inner products $\mathbf{G} = \mathbf{X}^T\mathbf{X}$:

$$\mathbf{G}_{ij} = \mathbf{x}_i \cdot \mathbf{x}_j$$

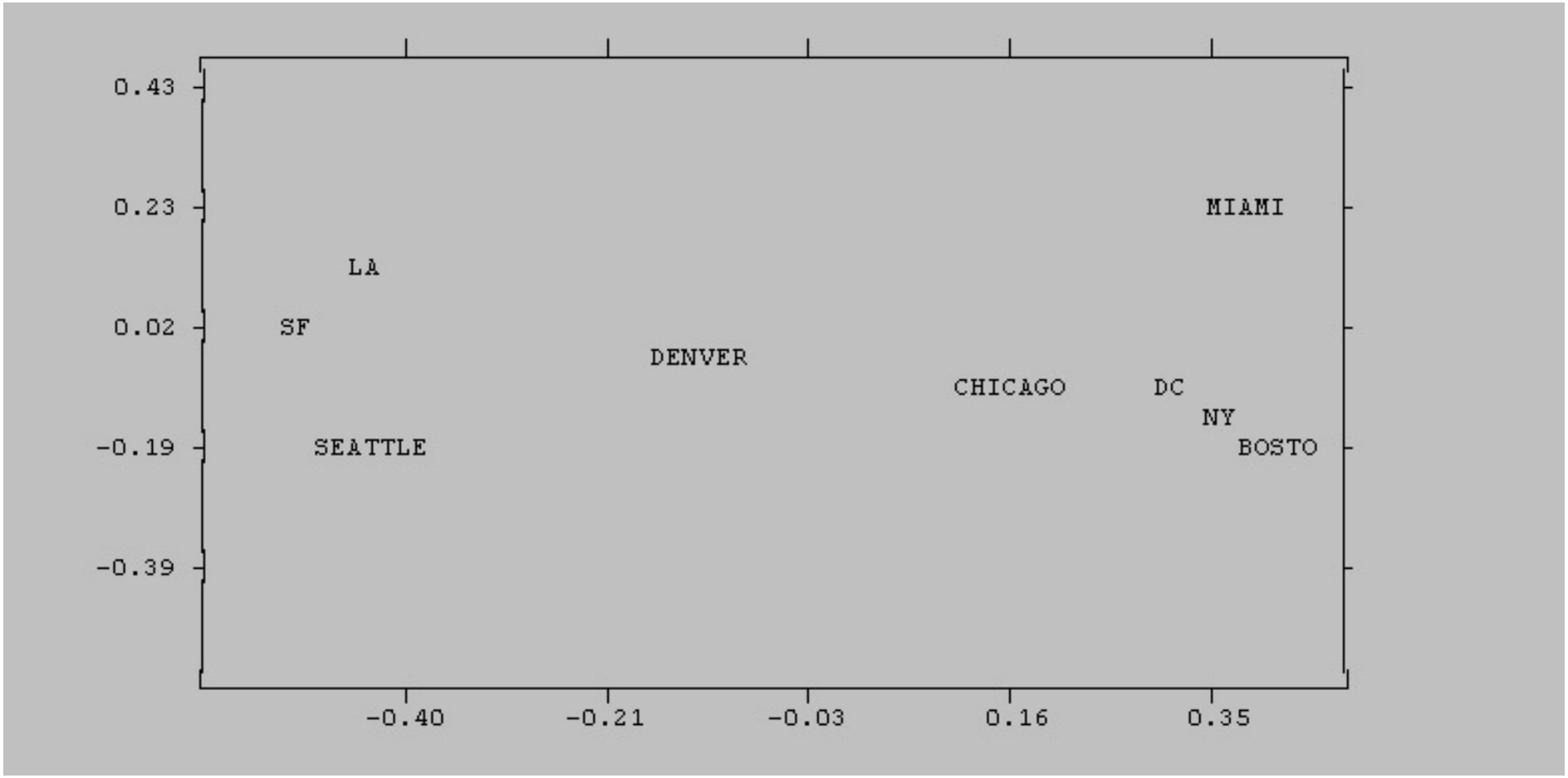
- The output projections are given by

$$\mathbf{z}_{i\alpha} = \sqrt{\lambda_\alpha}\mathbf{v}_{\alpha i},\ 1 \leq \alpha \leq k$$

where $\mathbf{v}_\alpha$ is the αth eigenvector of $\mathbf{G}$ and λ_α the corresponding e-value

MDS example

Input: inter-city distances



Multidimensional scaling

Note:

- Unlike PCA, MDS only gives projections for the training set; it does not give us a way to project a new data point
- We will need an "out-of-sample extension" to project new data points
- We'll come back to this later

Multidimensional scaling

MDS is based on the inner products between data points, but often we start with the matrix of Euclidean distances

$$\mathbf{F}_{ij} = ||\mathbf{x}_i - \mathbf{x}_j||^2 = ||\mathbf{x}_i||^2 - 2\mathbf{x}_i^T\mathbf{x}_j + ||\mathbf{x}_j||^2$$

If the $\mathbf{x}_i$ are zero-mean, we can convert the Euclidean distance matrix to a Gram matrix (inner-product matrix) by left- and right-multiplying by the *centering matrix*:

$$\mathbf{G} = -\frac{1}{2}(\mathbf{I} - \mathbf{u}\mathbf{u}^T)\mathbf{F}(\mathbf{I} - \mathbf{u}\mathbf{u}^T)$$

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In this case, the MDS projections $\mathbf{z}_i$ are the same as those of PCA:

- PCA directions are eigenvectors of $\mathbf{S} = \mathbf{X}\mathbf{X}^T$
- For each e-vector $\mathbf{w}_l$ of $\mathbf{S}$, there is a corresponding e-vector $\mathbf{v}_l = \mathbf{X}^T\mathbf{w}_l$ of $\mathbf{G} = \mathbf{X}^T\mathbf{X}$
- The first k vectors $\mathbf{v}_l$ give the projected data in both PCA and MDS